

Sobhan Salarian | Data Engineer

Sydney, NSW, Australia

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Profile

Data Engineer and Master of Data Science student with hands-on experience in Python and SQL for data cleaning, analysis and ETL pipeline development. Experienced in machine learning, data visualization, and NLP. Interested in building reliable data systems and translating data into actionable insights.

Education

Macquarie University <i>Master of Data Science</i>	Sydney, NSW <i>2025–Present</i>
Kermanshah University of Technology <i>Bachelor of Computer Engineering</i>	Kermanshah, Iran <i>2018–2023</i>
Adel High School of Isfahan University <i>High School (Mathematics Major)</i>	Isfahan, Iran <i>2014–2018</i>

Technical Skills

Programming: Python (Pandas, NumPy), SQL

Data Engineering: ETL pipelines, data ingestion, data transformation, batch processing, data modeling, pipeline monitoring and logging

Integration & Orchestration: Apache Airflow, Azure Data Factory (pipelines, triggers, linked services), REST API consumption

Cloud Platforms: AWS (S3, EC2), Microsoft Azure (Azure SQL, Blob Storage)

ML/NLP: scikit-learn, Hugging Face (Transformers/Datasets), PyTorch (training workflows), model evaluation (BLEU/chrF)

Data Analysis: Data cleaning, EDA, statistical analysis, feature engineering

Big Data: MapReduce (MrJob), MongoDB (PyMongo)

Databases: MySQL, MongoDB

Visualization: Power BI, Matplotlib, Bokeh

Tools: Git, GitHub, Jupyter, VS Code, IT change management awareness, technical documentation

Web: HTML, CSS

Awards & Achievements

- Awarded the highest mark in **COMP6179 – Statistical Technologies for Data Science** at Macquarie University (Session 2, 2025).

Professional Experience

President, Big Data Society ([Website](#))

2026–Present

- Lead a student data community founded by Prof. Amin Beheshti, bridging academia and industry collaboration.
- Build partnerships with industry professionals to explore real-world data challenges and applications.
- Organize workshops, networking sessions, and speaker events in **data science, big data, and AI**.
- Collaborate with university stakeholders and external partners to drive engagement and knowledge sharing.

IRISA Company

Isfahan, Iran

Data Science Intern

Jul–Sep 2022

- Cleaned and standardized datasets in **Python**, handling missing values, outliers, and inconsistencies.
- Loaded processed data into **MySQL**; collaborated with reporting team to clarify data requirements and communicate pipeline outputs to stakeholders.
- Documented data processing steps and performed quality checks to improve pipeline reliability; contributed to the internal knowledge base for ongoing operational reference.

Kish Language School

Isfahan, Iran

English Language Instructor

- Taught English to pre-intermediate and intermediate students aged 8–18 in classroom environments.
- Developed communication, presentation, and student engagement skills through lesson delivery and classroom management.
- Adapted teaching approaches for different age groups and learning abilities to improve participation and understanding.

Projects

*CRM Enhancement with AI-Powered Recommender System | **AI, Project Management, Risk Analysis***

- Designed a data-driven CRM solution using an **AI-based recommender system** to enhance customer engagement and personalization.
- Applied **Agile-Waterfall hybrid methodology** to manage project phases, deliverables, and milestones.
- Identified and mitigated key risks (**data quality, privacy, system integration**) using structured risk management strategies.
- Conducted **stakeholder analysis** and collaborated with cross-functional teams to align business requirements and ensure adoption.
- Defined **KPIs** targeting improvements in customer retention, conversion rates, and marketing ROI.

*Bitcoin Price Prediction using **Databricks** | **Machine Learning, Data Analysis** ([Certificate link](#))*

- Performed **exploratory data analysis (EDA)** and visualized Bitcoin market trends.
- Ingested data via **REST API** and preprocessed datasets for **machine learning pipelines**.
- Implemented a **regression model** in Databricks to predict Bitcoin price movements.

*Multilingual ED Triage Translation | **NLP, Machine Translation, Hugging Face***

2025

- Designed safe **Chinese↔English** translation for clinical **LEP** scenarios.
- Processed medical datasets and built **tokenization and validation pipelines**.
- Fine-tuned **NLLB-200** using **GPU training** with Hugging Face workflows.
- Evaluated models using **BLEU/chrF** and conducted error analysis.

*Spatial Data Analysis | **R-Trees, kNN, Skyline Queries***

2025

- Implemented an **R-Tree index** for efficient spatial querying.
- Developed **kNN** and **skyline algorithms** for location-based analysis.
- Benchmarked **multiple algorithms** for performance and scalability.
- **Presentation:** Google Drive link

*Big Data Weather Analysis | **MapReduce, MongoDB***

2025

- Processed large datasets using **MongoDB** and **MapReduce** with MrJob.
- Built **distributed data pipelines** for extreme weather detection and analytics.
- Documented **data workflows and system architecture**.

*Tennis Data Analysis | **Python, Pandas, Matplotlib***

2024

- Analyzed datasets using **Pandas/NumPy** to identify performance patterns.
- Processed parquet data and created **reproducible analysis notebooks**.
- Visualized insights using **Matplotlib**.

Flight Delay Analysis System | Python, Bokeh, Machine Learning

2023

- Analyzed delay patterns using **Pandas** and interactive visualizations in **Bokeh**.
- Developed a **recommendation model** for airline selection based on delay trends.
- Built **interactive dashboards** for data exploration.

Professional Certifications

Neo4j GraphAcademy

Neo4j & Generative AI Fundamentals

2025

- Covered **Generative AI fundamentals**, **LLMs**, and retrieval-augmented generation patterns (**RAG/GraphRAG**).
- Explored integrating **Neo4j** with generative AI applications and graph-based retrieval workflows.
- Credential: [Link](#)

Daneshkar Academy

Data Analysis Bootcamp

2024

- Covered statistics, SQL/MySQL, and data manipulation with **Pandas/NumPy**.
- Built dashboards and reports in **Power BI** and practiced Git/GitHub workflows.

Coursera

Python Programming for Everybody

2023–2024

- Completed Python fundamentals and intermediate concepts, including basic data analysis and database integration.

Languages

Persian (Farsi): Native

English: Professional Working Proficiency